First Order / Predicate logic

D = the set of students
↓
domain

1. "All CS students are smart".

$CS: D \to \{T, F\}$, $CS(x) = T$ if "x is a CS student"

$smart: D \to \{T, F\}$, $smart(x) = T$ if "x is smart"

$(\forall) x,\ cs(x) \to smart(x)$

2. "There is someone who studies at BBU and is smart"

$st: D \times U \to \{T, F\}$, $st(x, y) = T$ if "x studies at y"

U = the set of all universities, $BBU \in U$.

$(\exists x),\ (st(x, BBU) \wedge smart(x))$

3. "If x and y are nonnegative integers and $x > y$, then x^2 is greater than y^2".

$D = \mathbb{Z}$

predicate functions (returns T or F)

$mneg: D \to \{T, F\}$, $mneg(x) = T$ if $x \geq 0$

$gt: D \times D \to \{T, F\}$, $gt(x, y) = T$ if $x > y$

function symbol (return a symbol / obj.) ← $sq: D \to D$, $sq(x) = x^2$

$(\forall) x, y\ (mneg(x) \wedge mneg(y) \wedge gt(x, y) \to gt(sq(x), sq(y)))$

For the optional hw, work like this.

D = domain (the universe of birds)

Piky = constant of the universe

unary pred. symb: hb, rc, sb, lh

hb → hummingbird, rc → richly colored, sb → small bird, lh → lives on honey

$H_1: (\forall) x)\ hb(x) \to rc(x)$

→ "All hummingbirds are richly coloured."

H_2: $\neg(\exists x), (\neg sb(x) \wedge lh(x)) \equiv (\forall x)(\neg sb(x) \to \neg lh(x))$

↳ „No large birds live on honey."

H_3: $(\forall x)(\neg lh(x) \to \neg rc(x))$

↳ „Birds that do not live on honey are dull in color."

H_4: $hb(Piky)$

↳ „Piky is a hummingbird".

C_1: $(\exists x)\, lh(x)$ → „There is a bird which lives on honey"

C_2: $(\forall x)(hb(x) \to sb(x))$ → „All hummingbirds are small".

Building deductions

↳ Inferences rules:

Universal instatuation.

$H_1 \vdash_{\text{univ. inst.},\, x \leftarrow Piky} hb(Piky) \to rc(Piky) \quad (f5)$

$H_2 \vdash_{\text{univ. inst.},\, x \leftarrow Piky} \neg sb(Piky) \to \neg lh(Piky) \quad (f6)$

$H_3 \vdash_{\text{univ. inst.},\, x \leftarrow Piky} \neg lh(Piky) \to \neg rc(Piky) \quad (f7)$

H_4: $hb(Piky)$

$H_4, f5 \vdash_{mp} rc(Piky) \quad (f8)$

$f8, f7 \vdash_{mt} lh(Piky) \quad (f9)$

$f9 \vdash_{\text{exist general}} (\exists x)\, lh(x) \quad (C_1)$

↳ „There is a bird which lives on honey."

$H_1 \vdash_{\text{univ. inst.}} hb(x) \to rc(x) \quad (f10)$

$H_2 \vdash_{\text{univ. inst.}} \neg sb(x) \to \neg lh(x) \quad (f11)$

$H_3 \vdash_{\text{univ. inst.}} \neg lh(x) \to \neg rc(x) \quad (f12)$

$$f_{11}, f_{12} \xmapsto[\text{syllogism}]{} \neg sb(x) \to \neg rc(x) \quad (f_{13})$$

$$f_{13} \xmapsto[\text{mt}]{} rc(x) \to sb(x) \quad (f_{14})$$

$$f_{10}, f_{14} \xmapsto[\text{syllogism}]{} hb(x) \to sb(x) \quad (f_{15})$$

$$f_{15} \xmapsto[\text{univ. general}]{} (\forall x)(hb(x) \to sb(x)) \quad (C_2)$$

↳ "All hummingbirds are small."

When proving a theorem, make use of instatuation!
↓
inference rule.

Semantic approach of predicate logic

$$\boxed{\begin{array}{l} ((\forall)x)\, A(x) \equiv A(a) \wedge A(b) \\ \quad x \in \{a,b\} \\ (\exists x)\, A(x) \equiv A(a) \vee A(b) \\ \quad x \in \{a,b\} \end{array}}$$

EXPANSION LAWS.

Quantifiers interchanging laws

$$\left.\begin{array}{l} (\exists x)(\exists y)\, A(x,y) \equiv (\exists y)(\exists x)\, A(x,y) \\ (\forall x),(\forall y)\, A(x,y) \equiv (\forall y)(\forall x)\, A(x,y) \end{array}\right\} \text{Not applicable for } \exists + \forall.$$